NBA Awards Prediction

Introduction

This report details the methodology, implementation, and results of a predictive modeling project aimed at forecasting the end-of-season All-NBA and All-Rookie teams.

All-NBA and All-Rookie Teams

The All-NBA Team is an annual honor bestowed upon the best players in the National Basketball Association (NBA) following every regular season. Voted on by a global panel of sportswriters and broadcasters, the honor is divided into three distinct five-man lineups: the First, Second, and Third teams.

Similarly, the All-Rookie Team honors the top performing first-year players in the league. Voted on by NBA head coaches, this distinction is divided into two five-man lineups: the First and Second teams.

Metric

To evaluate the model's prediction performance, a custom, point-based scoring system is utilized.

Points are awarded per player based on the following criteria:

Exact Match: **10 points** (The player is placed in the correct team). Off by One: **8 points** (The player is placed one team above or below their actual result). Off by Two: **6 points** (The player is placed two teams above or below their actual result). Un-nominated: **0 points**

To further reward highly accurate models, cumulative bonus points are awarded for correctly predicting multiple players within the same specific exact lineup: 2 correct players: +5 points 3 correct players: +10 points 4 correct players: +20 points 5 correct players: +40 points

Maximum Possible Score & Normalization A perfect prediction for a single team (5 players at 10 points each, plus the 40-point completion bonus) yields exactly 90 points. Across the five target teams, a perfect sweep results in 450 points.

Approach

The predictive modeling approach is formulated as a *regression* problem, where the model is trained to predict a *vote share*. This is a `0-1` percentage value, which is used to create a ranking of players. The top 5 players are then selected as the predicted team for each category (All-NBA and All-Rookie).

Sparsity

One of the problems with the data is the sparsity of votes. Almost *all* players receive `0` votes, and only a small fraction of players receive any votes at all. This creates a highly imbalanced dataset, which can be challenging for traditional regression models to learn from effectively.

To tackle this issue, a **hurdle regression** approach is implemented. This involves first training a **binary classifier** to predict probabilities whether a player receives any votes (i.e., whether they are nominated or not). Then, for the a separate regression model is trained to predict their vote share. This two-step process helps to mitigate the effects of sparsity and allows the model to focus on learning from the relevant subset of data.

The comparison to a standard regression model is included in *Experiments* section below.

Code

The hurdle regression is implemented in `src/pipeline.py`

Data Acquisition

Despite available official nba stat apis, the task of gathering data was very painstaking.

To gather league statistics, the `nba_api` package was used [[link][(https://github.com/swar/nba_api). However, the package do not provide access to the voting data.

Both basic and advanced statistics were collected, ensuring they are taken from regular season, and not from the playoffs, as the voting is based on regular season performance.

Additionaly, column regarding team position in the final season ranking was added as an additional feature

The voting data was scraped from the basketball reference website [[link][(https://www.basketball-reference.com/awards/awards_2023.html). This process involved writing custom web scraping scripts to extract the relevant information from the HTML pages

Finally the data was cleaned and preprocessed in `R` as it is much more convenient and superior than `python` for tabular data manipulation and cleaning. The cleaned data was then exported to `dataset/final_ml_dataset.csv` file

Feature selection

The following features were selected for the model: 0. PTS_RANK: A team or player's league-wide ranking based on their average points scored per game. 0. PIE_RANK: A ranking based on the Player Impact Estimate (PIE), indicating an entity's overall statistical contribution compared to the rest of the league. 0. PLUS_MINUS_RANK: The league ranking of a player's or team's average point differential (Plus/Minus) while they are on the court. 0. W_PCT_RANK: A team's rank based on their win percentage during the regular season. 0. PTS: The average number of points scored per game. 0. REB: The average number of total rebounds (offensive plus defensive) collected per game. 0. AST: The average number of assists distributed per game, representing passes that directly lead to a scored basket. 0. STL: The average number of steals recorded per game by legally taking the ball away from an opponent. 0. BLK: The average number of blocked shots per game where a defensive player legally deflects a field goal attempt. 0. USG_PCT: Usage Percentage estimates the percentage of team plays used by a specific player while they are on the floor. 0. TS_PCT: True Shooting Percentage measures overall shooting efficiency by factoring in the value of two-point field goals, three-point field goals, and free throws. 0. PlayoffRank: A team's current standing or projected seed within their respective conference for playoff qualification. 0. WinPCT: The ratio of games won to total games played, representing the team's overall winning percentage. 0. TD3: The total number of triple-doubles achieved, which occurs when a player accumulates double digits in three of the five major statistical categories in a single game. 0. GP: The total number of games played by a player or team during the specified season. 0. IS_ROOKIE: A boolean or binary flag indicating whether a player is in their first active season in the NBA.

Feature standardization

Because absolute value features (e.g., points, rebounds) are not directly comparable across different seasons due to changes in the style of play and league dynamics, they were standardized using league-wide rankings. This approach allows for a more consistent comparison of player performance across different eras.

Code

1. Obtaining league statistics: `src/data/get_league_stats.py`
2. Scraping voting data: `src/data/scrape_voting_data.py`
3. Data cleaning and preprocessing: `src/data/preprocessing.R`

Problems

In reality there is not always 5 players per team For example in 1995-1996, there was a tie in the all-rookie nba voting, which resulted in 6 players being selected for the first team and only 4 players being selected for the second team.

All-Rookie Teams

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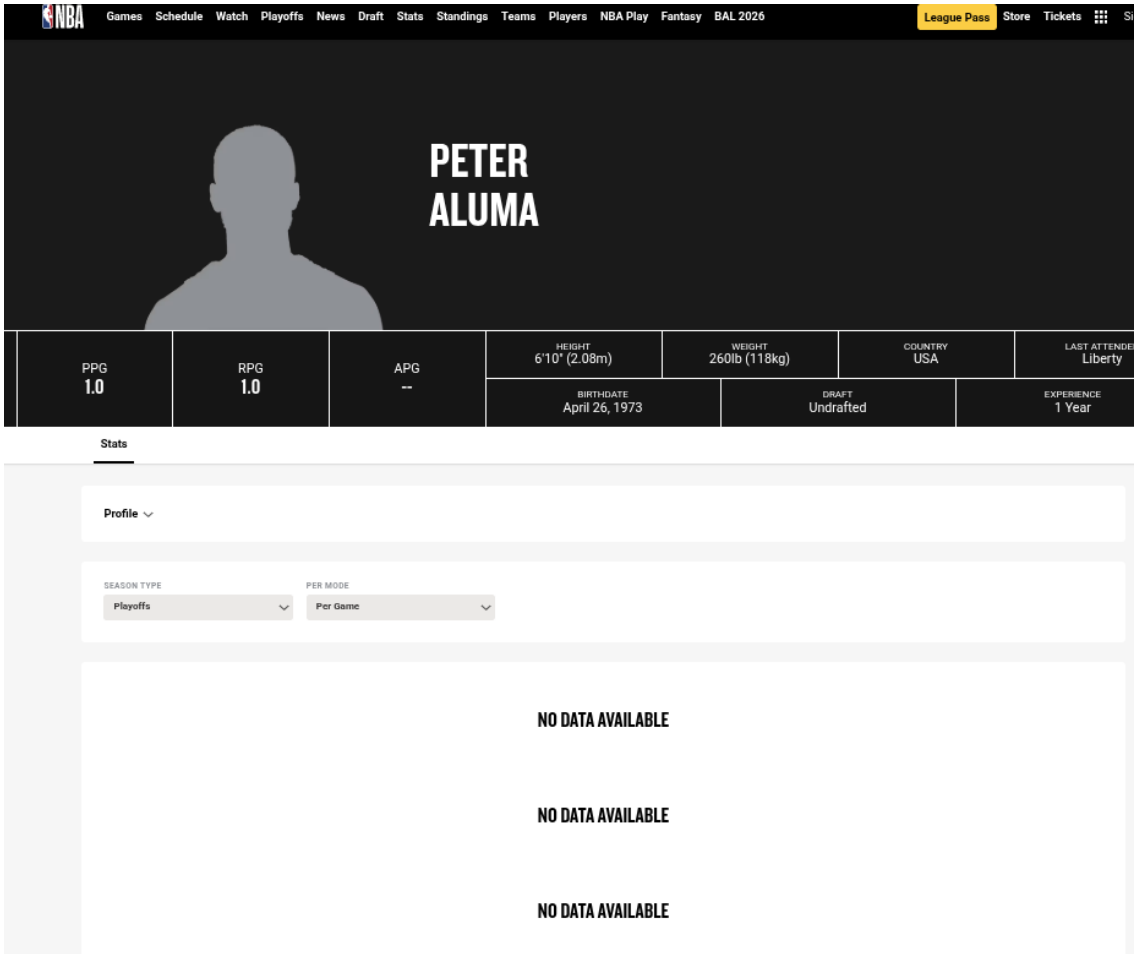
				Voting						Per Game							Shooting			Advanced	
# Tm	Player	Age	Tm	Pts Won	Pts Max	Share	1st Tm	2nd Tm	G	MP	PTS	TRB	AST	STL	BLK	FG%	3P%	FT%	WS	WS/48
1st	Damon Stoudamire	22	TOR	56	56	1.000	28	0	70	40.9	19.0	4.0	9.3	1.4	0.3	.426	.395	.797	4.3	.072
1st	Joe Smith	20	GSW	54	56	0.964	26	2	82	34.4	15.3	8.7	1.0	1.0	1.6	.458	.357	.773	6.8	.116
1st	Jerry Stackhouse	21	PHI	52	56	0.929	25	2	72	37.5	19.2	3.7	3.9	1.1	1.1	.414	.318	.747	1.2	.021
1st	Antonio McDyess	21	DEN	50	56	0.893	23	4	76	30.0	13.4	7.5	1.0	0.7	1.5	.485	.000	.683	3.6	.075
1st	Arvydas Sabonis	31	POR	41	56	0.732	17	7	73	23.8	14.5	8.1	1.8	0.9	1.1	.545	.375	.757	8.4	.233
1st	Michael Finley	22	PHO	41	56	0.732	15	11	82	39.2	15.0	4.6	3.5	1.0	0.4	.476	.328	.749	6.3	.095
2nd	Kevin Garnett	19	MIN	35	56	0.625	7	21	80	28.7	10.4	6.3	1.8	1.1	1.6	.491	.286	.705	4.4	.092
2nd	Bryant Reeves	22	VAN	28	56	0.500	2	24	77	31.9	13.3	7.4	1.4	0.6	0.7	.457	.000	.732	2.3	.045
2nd	Brent Barry	24	LAC	21	56	0.375	1	19	79	24.0	10.1	2.1	2.9	1.2	0.3	.474	.416	.810	4.9	.124
2nd	Rasheed Wallace	21	WSB	15	56	0.268	1	13	65	27.5	10.1	4.7	1.3	0.6	0.8	.487	.329	.650	2.3	.063
2nd	Tyus Edney	22	SAC	15	56	0.268	0	15	80	31.0	10.8	2.5	6.1	1.1	0.0	.412	.368	.782	3.3	.064
ORV	Kurt Thomas	23	MIA	14	56	0.250	0	14	74	22.4	9.0	5.9	0.6	0.6	0.5	.501	.000	.663	3.8	.110
ORV	Eric Williams	23	BOS	9	56	0.161	0	9	64	23.0	10.7	3.4	1.1	0.9	0.2	.441	.300	.671	1.6	.052
ORV	Alan Henderson	23	ATL	1	56	0.018	0	1	79	17.9	6.4	4.5	0.6	0.6	0.5	.442	.000	.595	2.0	.069
ORV	Don Reid	22	DET	1	56	0.018	0	1	69	14.4	3.8	2.9	0.2	0.7	0.6	.567		.662	2.9	.141
ORV	Bob Sura	22	CLE	1	56	0.018	0	1	79	14.6	5.3	1.7	2.9	0.7	0.3	.411	.346	.702	1.6	.069

All-Rookie Teams

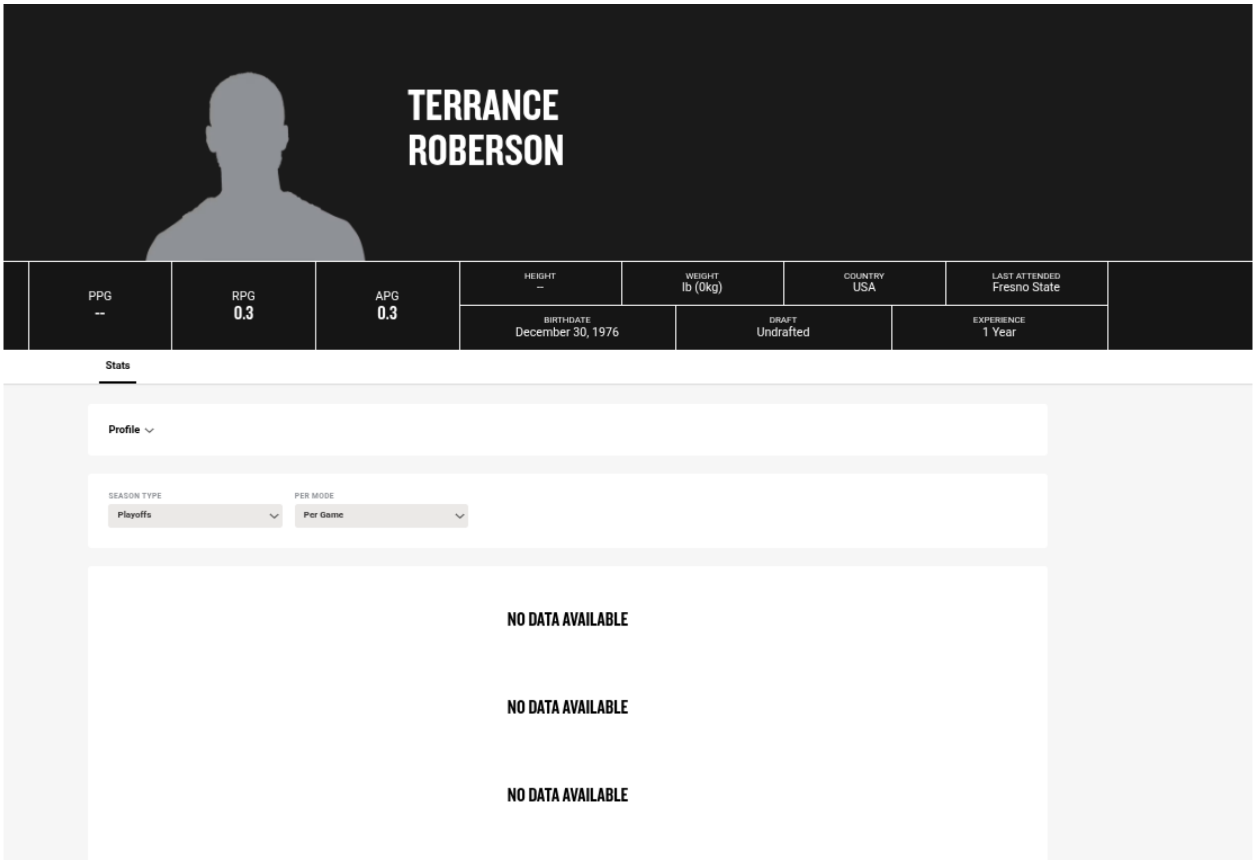
Share & Export ▾ Glossary

				Voting						Per Game					
# Tm	Player	Age	Tm	Pts Won	Pts Max	Share	1st Tm	2nd Tm	G	MP	PTS	TRB	AST	STL	BLK
1st	Brandon Roy	22	POR	58	58	1.000	29	0	57	35.4	16.8	4.4	4.0	1.2	0.2
1st	Andrea Bargnani	21	TOR	57	58	0.983	28	1	65	25.1	11.6	3.9	0.8	0.5	0.8
1st	Randy Foye	23	MIN	48	58	0.828	21	6	82	22.9	10.1	2.7	2.8	0.6	0.
1st	Rudy Gay	20	MEM	39	58	0.672	12	15	78	27.0	10.8	4.5	1.3	0.9	0.9
1st	LaMarcus Aldridge	21	POR	37	58	0.638	13	11	63	22.1	9.0	5.0	0.4	0.3	1.2
1st	Jorge Garbajosa	29	TOR	37	58	0.638	13	11	67	28.5	8.5	4.9	1.9	1.2	0.2
2nd	Paul Millsap	21	UTA	36	58	0.621	10	16	82	18.0	6.8	5.2	0.8	0.8	0.9
2nd	Adam Morrison	22	CHA	35	58	0.603	12	11	78	29.8	11.8	2.9	2.1	0.4	0.
2nd	Tyrus Thomas	20	CHI	26	58	0.448	5	16	72	13.4	5.2	3.7	0.6	0.6	1.
2nd	Craig Smith	23	MIN	21	58	0.362	1	19	82	18.7	7.4	5.1	0.6	0.6	0.2
2nd	Walter Herrmann	27	CHA	10	58	0.172	1	8	48	19.5	9.2	2.9	0.5	0.4	0.
2nd	Rajon Rondo	20	BOS	10	58	0.172	1	8	78	23.5	6.4	3.7	3.8	1.6	0.
2nd	Marcus Williams	21	NJN	10	58	0.172	1	8	79	16.6	6.8	2.1	3.3	0.4	0.
ORV	Tarence Kinsey	22	MEM	6	58	0.103	1	4	48	20.1	7.7	2.0	0.9	1.1	0.
ORV	Rodney Carney	22	PHI	5	58	0.086	1	3	67	17.4	6.6	1.9	0.4	0.6	0.
ORV	Renaldo Balkman	22	NYK	5	58	0.086	0	5	68	15.6	4.9	4.3	0.6	0.8	0.
ORV	Shelden Williams	23	ATL	4	58	0.069	0	4	81	18.7	5.5	5.4	0.5	0.6	0.5
ORV	Josh Boone	22	NJN	1	58	0.017	0	1	61	11.0	4.2	2.9	0.2	0.2	0.
ORV	Jordan Farmar	20	LAL	1	58	0.017	0	1	72	15.1	4.4	1.7	1.9	0.6	0.
ORV	Daniel Gibson	20	CLE	1	58	0.017	0	1	60	16.5	4.6	1.5	1.2	0.4	0.
ORV	Allan Ray	22	BOS	1	58	0.017	0	1	47	15.1	6.2	1.5	0.9	0.4	0.
ORV	Sergio Rodríguez	20	POR	1	58	0.017	0	1	67	12.9	3.7	1.4	3.3	0.5	0.
ORV	Thabo Sefolosha	22	CHI	1	58	0.017	0	1	71	12.2	3.6	2.2	0.8	0.5	0.2

The same situation happened in 2007



Missing data TODO



Missmatch in names between stats and voting data The scraped thata do not contain `PLAYER_ID` so it had to be obtained by matching the player names in the voting data with the player names in the stats data.

Sometimes, the names do not match exactly, so different heuristics were implemented including removing utf-8 characters, matching only the last name etc.

All heuristics matches were logged as warnings and validated manually.

Pipeline

The whole process was implemented in `sklearn` pipeline, which includes the following steps: 1. Selecting appropriate features 2. Regularization of the appropriate features 3. Predicting probabilities of being nominated using a binary classifier 4. Predicting vote share using a regression model 5. Scaling the predicted vote share by the predicted probabilities of being nominated

Experiments

I wanted to check the following:

1. How does excluding old seasons affect the performance of the model? In the past the game were different, and the voting criteria change as well

2. Is the classification step neccessary, if so how different classification models perform?
3. How do different regression models perform?

Validation

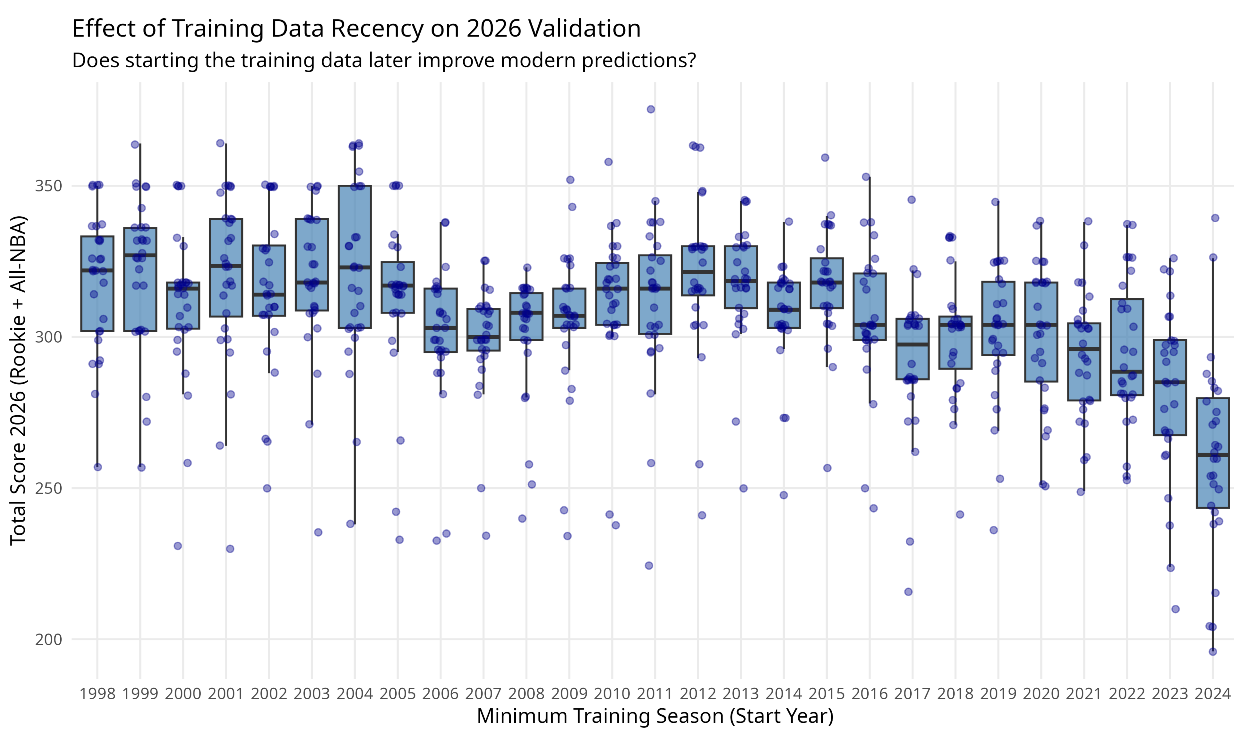
The validation was performed on the 2026 season, which is the most recent season for which the voting data is available. The model was trained on all previous seasons, and then used to predict the All-NBA and All-Rookie teams for the 2026 season. The predicted teams were then compared to the actual teams using the scoring system described in the *Metric* section above.

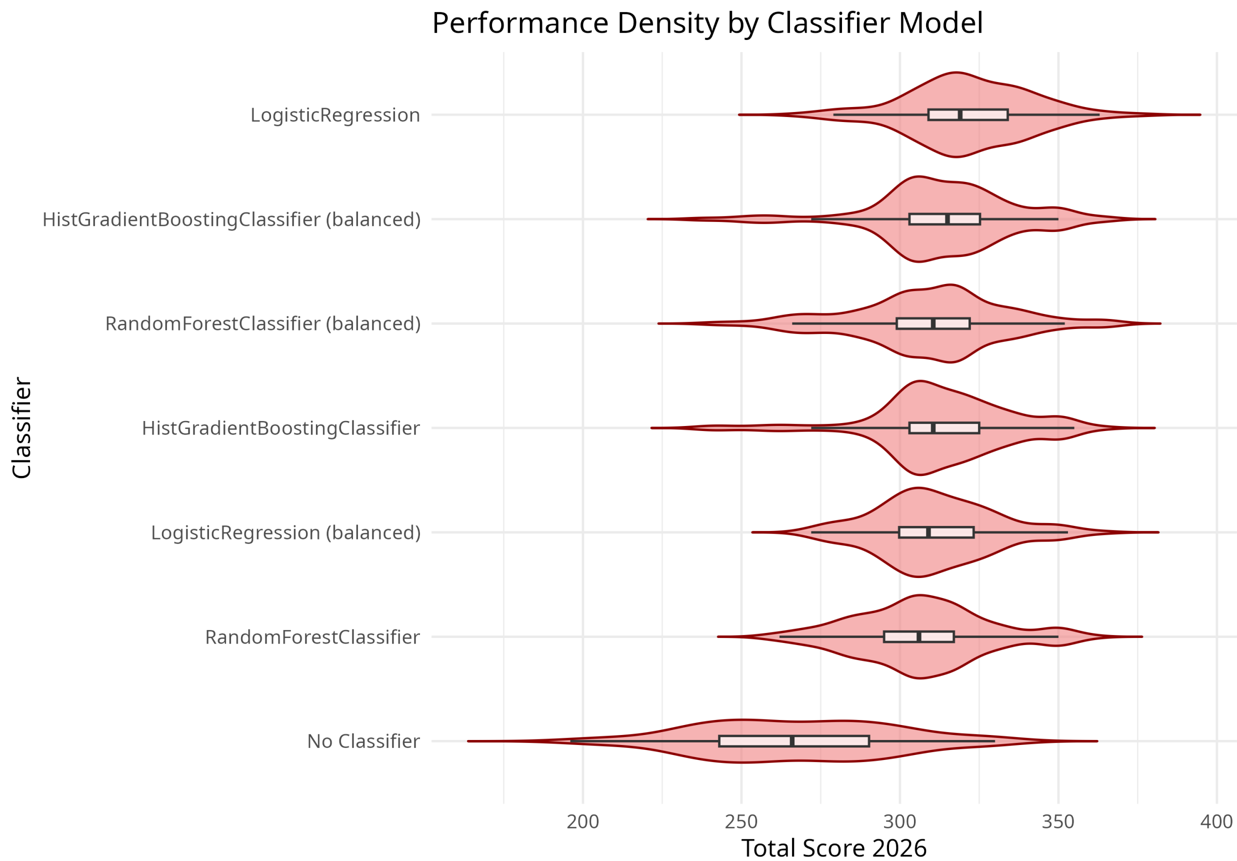
Probably this is not the most reliable validation strategy, but unfortunately, the rules for selecting the All-NBA and All-Rookie teams changed in 2023, which makes it difficult to use older seasons for validation. The new rules eliminate the requirement for diverse positions across the 5 players in the same team, as well as the recent minimum games played requirement, which significantly changes the voting dynamics and criteria.

Results

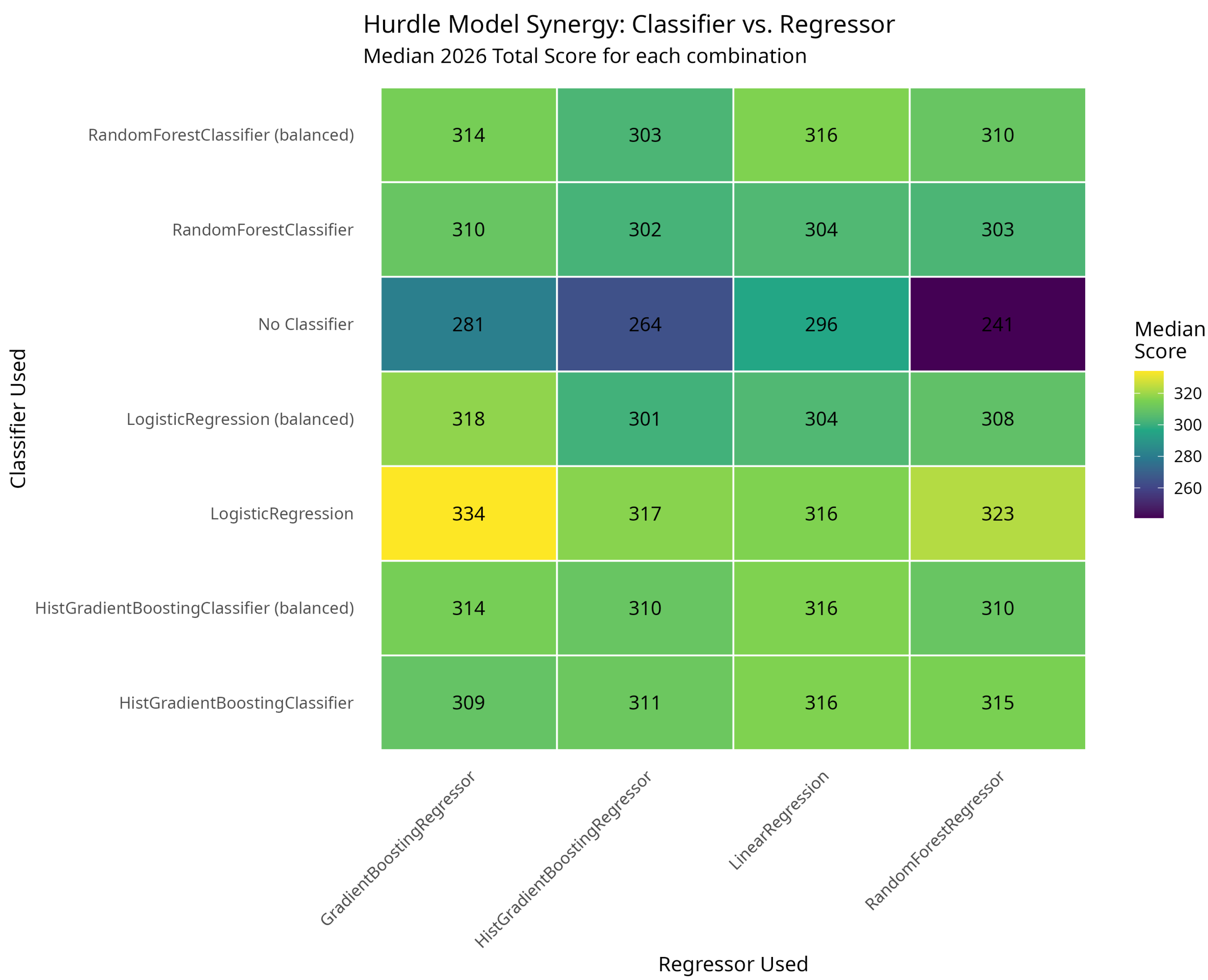
The best score 375 for 2026 season was achieved by the Gradient Boosting Regressor with the classification by Logistic Regression.

Analysis




Performance Density by Classifier Model
LogisticRegression
HistGradientBoostingClassifier (balanced)
RandomForestClassifier (balanced)
HistGradientBoostingClassifier
LogisticRegression (balanced)
RandomForestClassifier
No Classifier
Classifier
200
250
300
350
400
Total Score 2026


Performance Density by Regression Model
LinearRegression
GradientBoostingRegressor
RandomForestRegressor
HistGradientBoostingRegressor
Regressor
200
250
300
350
400
Total Score 2026


Hurdle Model Synergy: Classifier vs. Regressor
Median 2026 Total Score for each combination
Classifier Used
RandomForestClassifier (balanced)
RandomForestClassifier
No Classifier
LogisticRegression (balanced)
LogisticRegression
HistGradientBoostingClassifier (balanced)
HistGradientBoostingClassifier
314
303
316
310
310
302
304
303
281
264
296
241
318
301
304
308
334
317
316
323
314
310
316
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309
311
316
315
GradientBoostingRegressor
HistGradientBoostingRegressor
LinearRegression
RandomForestRegressor
Regressor Used
Median
Score
320
300
280
260

Possible improvements

1. Acknowledgin the rules change in 2023

The rules change in 2023 significantly affects the voting dynamics and criteria for selecting the All-NBA and All-Rookie teams. One of the changes is dropping the requirement for a concrete number of players from each position (e.g., guards, forwards, centers) in the All-NBA teams. Another change is the minimum games played requirement

2. Smarter feature selection

The current feature selection is based on domain knowledge and intuition. However, a more systematic approach could be implemented, such as using feature importance from a tree-based model or using a feature selection algorithm.

3. Hyperparameter tuning

The current model is trained with default hyperparameters. A more thorough approach would involve using techniques such as grid search or random search to find the optimal hyperparameters for both the classification and regression models.

4. Better validation strategy

The current validation is performed on a single season. Theoreticaly, a more robust approach would involve a cross-validation strategy, although because of the 2023 rule, which eliminates requirement for diverse positions across the 5 players in the same team, as well as the recent minimum games played requirement, it is hard to tell haw to acquire a real validation score.

5. Statistical significance testing

For the comparison results to be actually meaningful, it would be important to perform statistical significance testing to determine whether the differences in performance between different models and approaches are statistically significant or if they could be due to random chance.